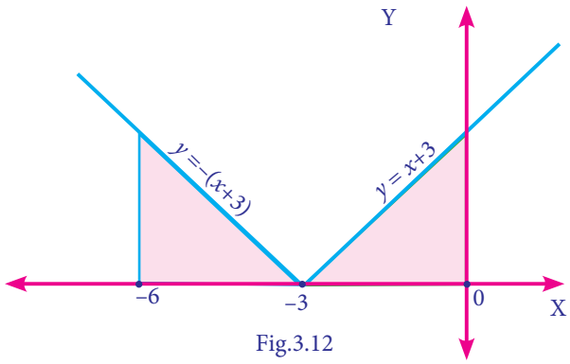


Fig.3.12

$$= \int_{-6}^{-3} y\,dx + \int_{-3}^{0} y dx$$

$$= \int_{-6}^{-3} -(x+3)dx + \int_{-3}^{0} (x+3)dx$$

$$= -\left[\frac{(x+3)^2}{2}\right]_{-6}^{-3} + \left[\frac{(x+3)^2}{2}\right]_{-3}^{0}$$

$$= -\left[0-\frac{9}{2}\right] + \left[\frac{9}{2}-0\right] = 9 \text{ sq. units}$$

Example 3.7

Using integration find the area of the circle whose center is at the origin and the radius is0 a units.

Solution

Equation of the required circle is $x^2 + y^2 = a^2$ (1)

put $y = 0$, $x^2 = a^2$

$\Rightarrow$ $x = \pm a$

Since equation (1) is symmetrical about both the axes

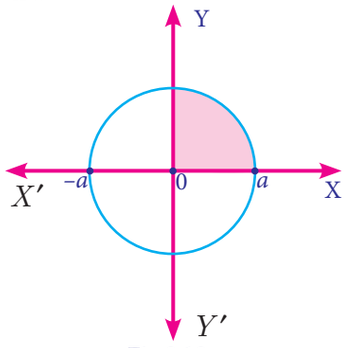


Fig.3.13

The required area = 4 [Area in the first quadrant between the limit 0 and a.]

$$= 4\int_0^a y\,dx$$

$$= 4\int_0^a \sqrt{a^2 - x^2}\,dx = 4\left[\frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]_0^a$$

$$= 4\left[0 + \frac{a^2}{2}\sin^{-1}\left(\frac{a}{a}\right)\right] = 4\left(\frac{a^2}{2}\sin^{-1}(1)\right) = 4.\frac{a^2}{2}\frac{\pi}{2}$$

$$= \pi a^2 \text{ sq. units}$$

Example 3.8

Using integration find the area of the region bounded between the line $x = 4$ and the parabola $y^2 = 16x$.

Solution:

The equation $y^2 = 16x$ represents a parabola (Open rightward)

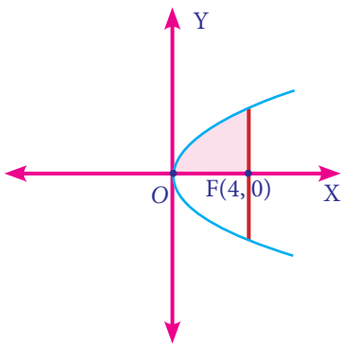


Fig.3.14

Required Area $= 2\int_0^a y dx$

$$= 2\int_0^4 \sqrt{16x}\,dx$$

$$= 8\int_0^4 x^{\frac{1}{2}}\,dx = 8\left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^4 = \frac{16}{3}\left((4)^{\frac{3}{2}}\right) = \frac{128}{3.} \text{ sq. units}$$

Exercise 3.1

1. Using Integration, find the area of the region bounded the line $2y + x = 8$, the x axis and the lines $x = 2$, $x = 4$.

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